

Optimization of Small Object Detection in UAV Aerial Imagery

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1. Introduction

- UAV Imagery Detection Challenges:
 - Extreme Target Scales: Critical ground targets (pedestrians/vehicles) often span less than 32x32 pixels with sparse features.
 - Downsampling Feature Loss: Mainstream detectors use standard strided-convolutions that discard tiny object signatures in deep layers.
 - Resolution Dilemma: Direct image downscaling destroys fine details, while native full-resolution processing creates excessive computing costs.
- Research Objective: Propose an optimized YOLOv11-based detection framework, integrated high-resolution processing strategies (Slicing & SAHI), to improve the small objects detection in UAV scenarios.

2. Proposed Architectural Optimizations

- BiFPN Neck Upgrade: Replaces YOLOv11 native PANet with a Bidirectional Feature Pyramid Network (BiFPN) for dynamic cross-scale feature integration.
 - Dynamic Feature Weights: Utilizes a learnable weight mechanism to automatically balance multi-resolution input layers.
 - Scale Migration (P2 Head): Removes the low-resolution P5 layer and introduces a shallow P2 layer head for rich geometric detail.

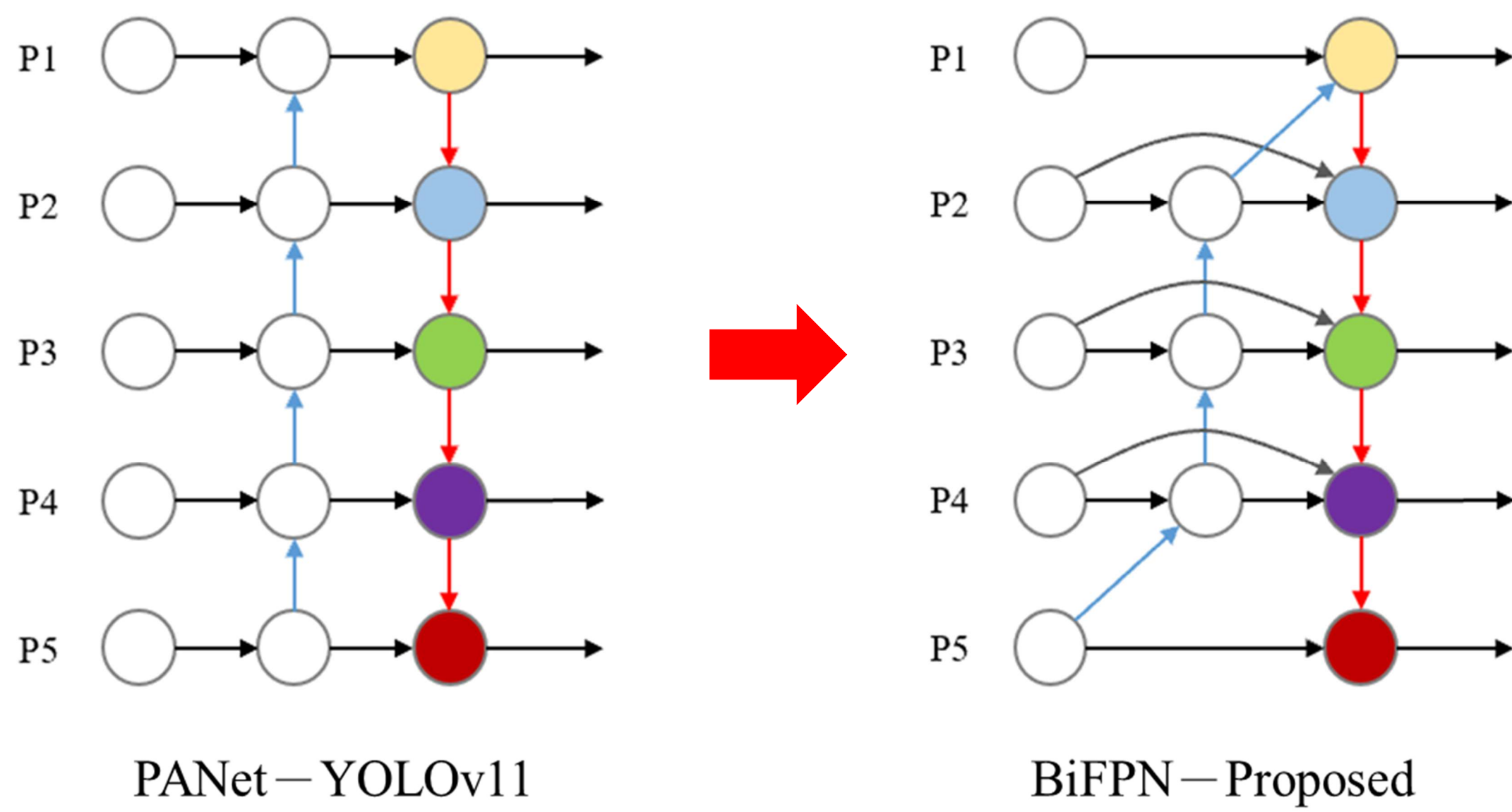


Figure 1. Two Neck Structures Comparison

- SPD-Conv Integration: Swaps out standard strided convolutions in early backbone stages for Space-to-Depth Convolutions.
 - Lossless Downsampling: Slices spatial matrices directly into the channel dimension to completely eliminate feature collapse.

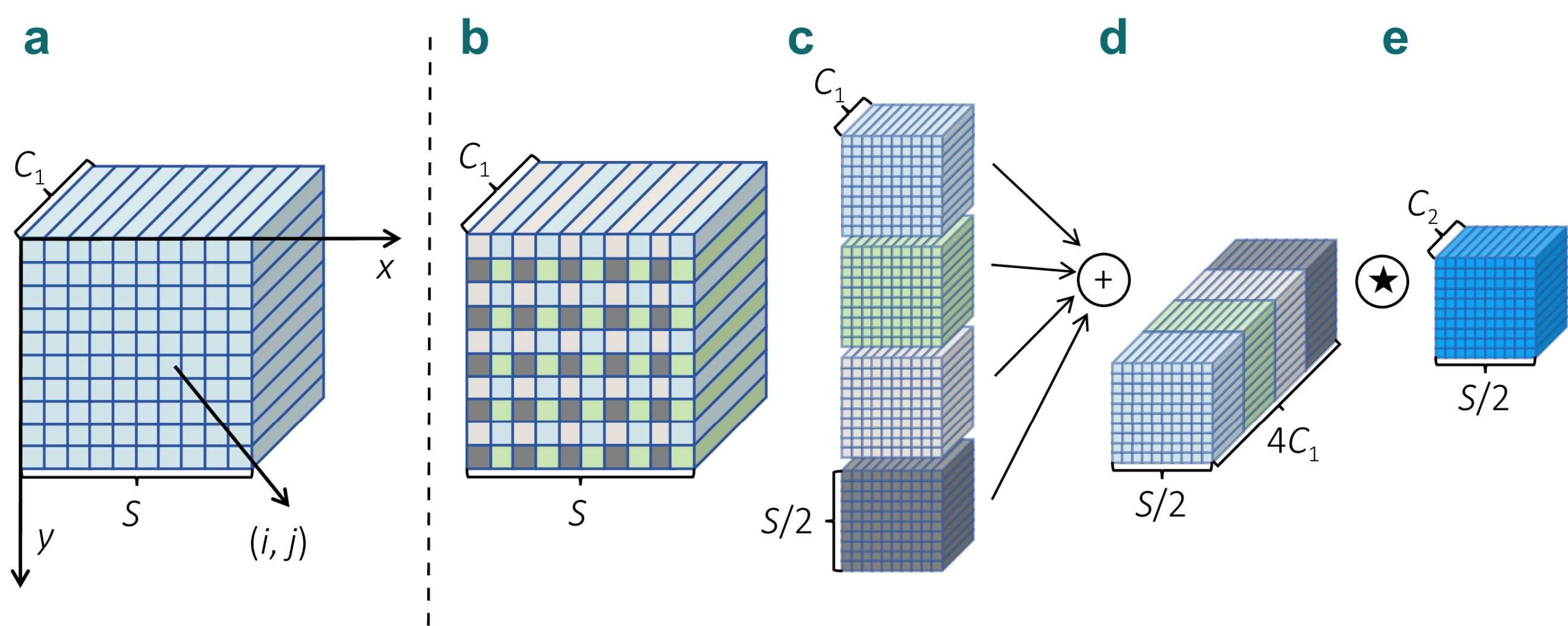


Figure 2. SPD-Conv Mechanism

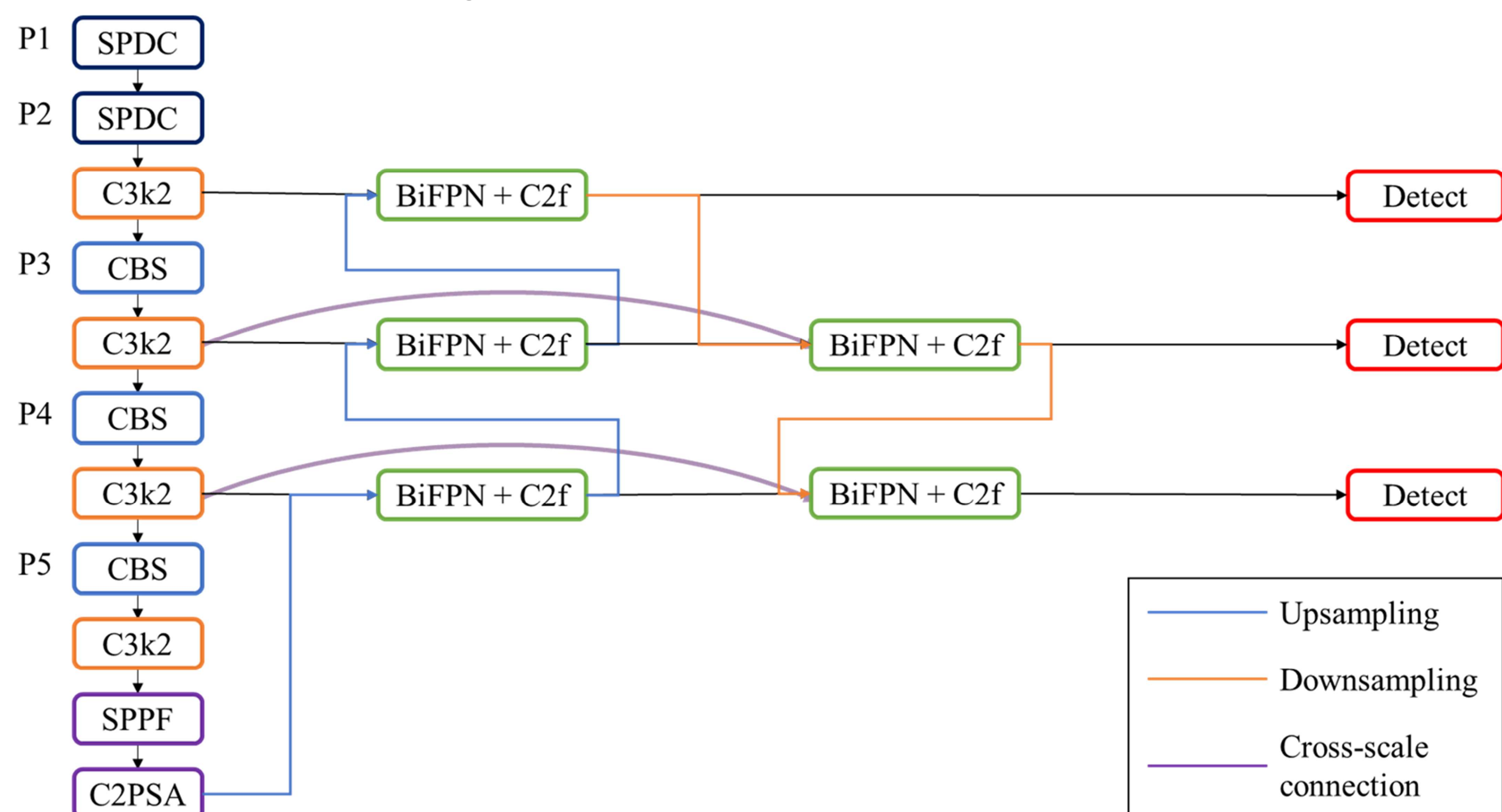


Figure 3. Proposed Model Structure

3. High-Resolution Strategy

- Employs parallel optimization during both training and evaluation phases without image distortion.

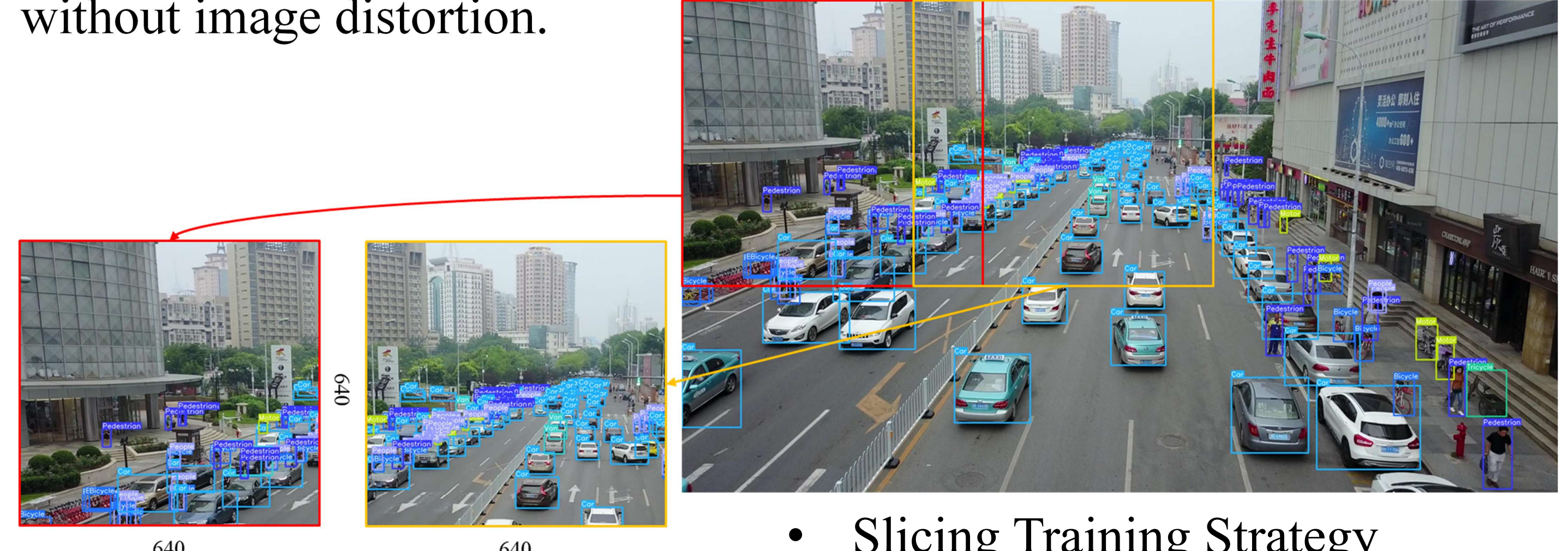


Figure 4. Data Slicing

- SAHI Detection Pipeline
 - Lossless Scan: Deploys SAHI using 640x640 windows to prevent pixel details from vanishing during full-frame downscaling.
 - Global Mapping: Runs batch inference on sub-patches and maps local coordinates back onto the native high-resolution canvas.
- Slicing Training Strategy
 - Overlap Patching: Slices raw images into 640x640 tiles with 20% overlap to prevent target truncation at borders.
 - Scale Filtering: Filters redundant layouts (>50% similarity) by prioritizing patches with smaller average target scales.

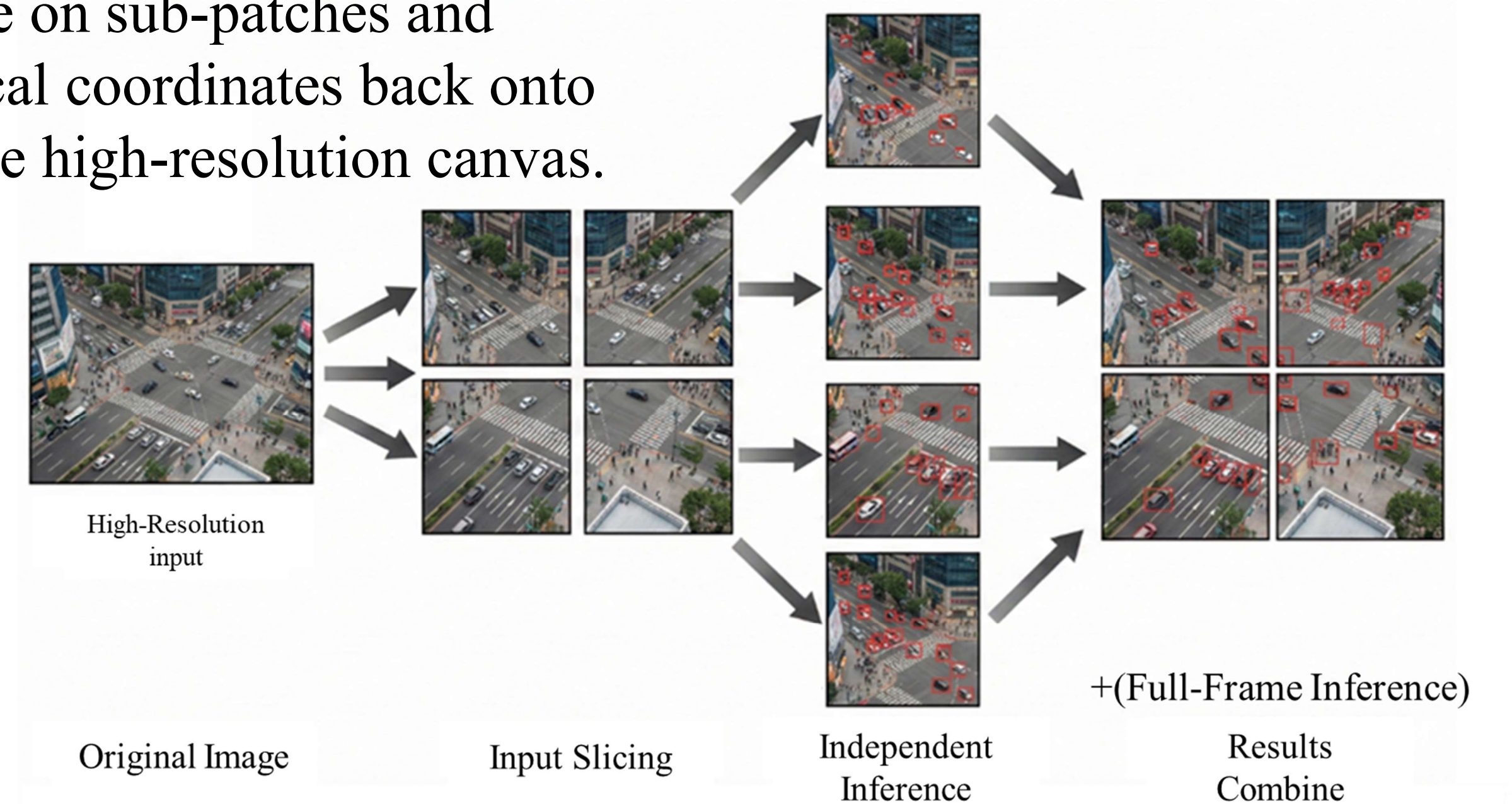


Figure 5. SAHI Work Flow

4. Experimental Results

- Dataset: Evaluated on VisDrone-DET2019.
- Proposed Model Boost: Reaches 50.40% mAP@50, outperforming the native baseline's 40.31% with a lightweight 12.02M parameter size.
- SAHI Pipeline Victory: Elevates final mAP@50 to a SOTA-level 61.82%, significantly mitigating false negatives.
- High Parameter Efficiency: Achieves detection accuracy comparable to heavy models (DRONE-YOLO) using only 1/6th of their parameter scale.

Model	mAP@50 (%)	mAP@50-95 (%)	Params (M)	GFLOPs
Faster R-CNN	37.2	21.9	41.16	292.8
YOLOv11s (Baseline)	40.31	23.69	9.43	21.6
DRONE-YOLO	51.3	31.9	76.20	-
Ours (Full Frame)	50.40	30.72	12.02	85.2
Ours (with SAHI)	61.82	37.67	12.02	85.2

Table 1. Experimental Results

5. Conclusion

- This study presents an optimized YOLOv11-s framework integrating BiFPN, SPD-Conv, and SAHI for UAV small object detection.
- Evaluated on the VisDrone benchmark, it achieves a definitive 61.82% mAP@50 with a lightweight 12.02M parameter footprint, matching heavy SOTA models like DRONE-YOLO at just one-sixth the size.
- Operating at 9.2 FPS, the throughput satisfies industrial quasi-real-time monitoring standards, demonstrating an exceptional balance between accuracy and efficiency for real-world aerial surveillance.